

TROUBLESHOOTING AND HELP DESK

TASK: 1

Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Removal and Reseating – Practical Report

1. Shut down the computer completely and disconnected the power cable/charger.
2. Removed the laptop back cover or opened the desktop cabinet using a screwdriver.
3. Located the **RAM slots** on the motherboard.
4. Before touching the RAM, discharged static electricity by touching a metal part of the computer case.
5. Pressed the locking clips on both sides of the RAM slot outward.
6. Carefully removed the RAM module by holding it from its edges without touching the gold contacts.
7. Inspected the RAM and slot for dust, damage, or loose connections.
8. Aligned the notch on the RAM module with the key in the RAM slot.
9. Inserted the RAM evenly into the slot and pressed down until the locking clips clicked into place.
10. Reconnected the power supply and switched on the computer.
11. The computer started normally, and the RAM was detected by the operating system.

Issues Observed:

No major issues were observed during the process. The RAM module was slightly difficult to remove initially because it was firmly secured by the locking clips. After reseating, the computer booted normally and the installed RAM was recognized correctly.

Result:

The RAM was successfully removed and re-inserted. The computer worked normally after reseating the RAM.

TASK: 2

Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

GPU Troubleshooting Checklist:

1. **Identify the Problem:** First, check the symptoms of the faulty GPU, such as no display, screen flickering, graphical glitches, crashes, or low gaming performance.
2. **Check Physical Connections:** Turn off the computer and check whether the GPU is properly inserted into the PCIe slot. Also, make sure the GPU power cables are connected properly.
3. **Check the Display Cable:** Make sure the HDMI or DisplayPort cable is securely connected to the GPU. Try another cable or monitor if the display is still not working.
4. **Check GPU Drivers:** Open **Device Manager** → **Display Adapters** and check whether the GPU is detected. Update or reinstall the graphics driver if necessary.
5. **Check GPU Cooling:** Check whether the GPU fans are working and whether the card is overheating. Remove dust and ensure proper airflow inside the cabinet.
6. **Test the GPU:** Run a game or graphics test to check for screen artifacts, crashes, freezing, or abnormal performance.
7. **Confirm the Fix:** Restart the PC and test the GPU again with games or other graphics applications. If the display works correctly, the GPU is detected, temperatures are normal, and there are no crashes or graphical problems, the issue is considered fixed.

TASK: 3

Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Storage Drive Unplugging and Reconnecting – Practical Report

Objective: To simulate a storage problem by disconnecting and reconnecting an SSD/HDD and observe the BIOS indicators and error messages.

Procedure and Observation: First, I shut down the test computer completely and disconnected the power supply. I carefully disconnected the SATA cable from the SSD/HDD and then started the computer. The BIOS did not detect the storage drive, and the system displayed a message such as **“No Boot Device Found”** or **“Operating System Not Found.”** In the BIOS storage or boot menu, the disconnected drive was also missing from the list. I then shut down the computer again and reconnected the SATA/data and power cables securely. After restarting the computer, the BIOS detected the storage drive, and the system booted normally into Windows.

Before Reconnecting:

- BIOS: Storage drive not detected.
- Boot screen: **“No Boot Device Found”** / **“Operating System Not Found.”**
- Windows: Could not start if the disconnected drive contained the operating system.

After Reconnecting:

- BIOS: SSD/HDD detected correctly.
- Boot menu: Storage drive appeared as a boot device.
- Windows: Started normally and the drive was accessible.

Conclusion: The test showed that a loose or disconnected SATA/NVMe drive can prevent the computer from detecting the storage device and may cause boot errors. Reconnecting the drive properly restored detection and normal system operation.

TASK: 4

Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Troubleshooting Random PC Restarts

If a PC randomly restarts, I would check the **RAM first**. Faulty, loose, or incompatible RAM can cause system crashes, freezing, blue-screen errors, and unexpected restarts. RAM is also easy to remove, reseal, and test.

Diagnostic Steps:

1. Shut down the PC and disconnect the power cable.
2. Open the computer case and locate the RAM modules.
3. Check whether the RAM is properly inserted into the RAM slots.
4. Remove and carefully reseal the RAM module.
5. If there are multiple RAM modules, test them one at a time to identify a faulty module.
6. Run **Windows Memory Diagnostic** to check for memory errors.
7. If errors are detected, replace the faulty RAM module.
8. Restart the PC and run games or other applications to confirm that the random restarting problem has been resolved.

Conclusion: I would check the RAM first because a faulty or poorly connected RAM module can cause random restarts. Reseating and testing the RAM can help identify and solve the problem.

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